Chemistry Exam Language Bank - Say It Like the Mark Scheme (9CH0)

Oct 15, 2026 · @ET

Purpose: Jody understands the chemistry, but Chemistry mark schemes award marks for exact definitions, named forces and bonds, and a clear chain of reasoning. This bank gives her the precise wording the mark scheme looks for, topic by topic, so she can learn to answer in that language.

用法簡述： 每個主題用三欄對照：概念、「自己的話」答法（失分）、「考試用語」答法（得分）。先蓋住右欄自己寫，再對照、劃出漏掉的術語。

How marks are awarded in Chemistry

- **Definitions are word-for-word.** "Ionisation energy" needs *one mole*, *gaseous atoms*, *1+ ions* and *electrons*. Leaving out a key phrase loses the mark.
- **Name the force or bond.** "Stronger forces" scores nothing; "stronger London forces between larger molecules" does. Say whether it is a covalent bond, ionic bond, hydrogen bond or London force, and whether it is *within* or *between* molecules.
- **Explain with a chain.** Size or charge, then attraction, then energy needed, then the observed property. Examiners tick each link.
- **Say what breaks.** For boiling and melting, state which bonds or forces are overcome, and that covalent bonds in simple molecules are not broken.
- **Calculations earn marks for method.** Show the formula, the substitution with units and the answer to a sensible number of significant figures. An error carried forward can still earn later marks.
- **Mechanisms are marked arrow by arrow.** An arrow must start on a lone pair, a bond or a negative charge and end on an atom or a bond.

How to use this bank

1. Read one row. Cover the right-hand column and write your own answer to the concept.
2. Uncover the exam-language answer. Underline every key term you missed.
3. Close the page and rewrite the answer using those terms. Repeat on a later day until the terms come without prompting.

The "own words" column shows typical answers that lose marks. After the mock paper is marked, replace them with Jody's own wording, which makes the contrast personal.

Command words: what each one asks for

Command word	What to write	Mark logic
State, Give, Name	One short fact or term, no explanation	1 mark for the correct term
Define	The accepted definition, with every key phrase	Each key phrase is a mark point
Describe	What happens, in order, with no reasons. For observations: the colour or state before and after	A mark for each observation
Explain	Why it happens: a point, then the chemical reason, then the consequence	Marks for the reason and the link, not the description
Suggest	Apply known chemistry to an unfamiliar situation	Marks for a sensible idea using correct chemistry
Compare	Similarities and differences, each stated for both items	Marks only when both sides are given
Calculate	Formula, substitution, answer with units	Method mark and answer mark
Deduce, Predict	A conclusion drawn from the data given, with reasoning if asked	Marks for a conclusion that follows from the data
Evaluate, Discuss	Advantages and disadvantages, then a reasoned judgement	Marks for balance and a justified conclusion
Show that	Every step, ending with the value given	The final value is given, so the working is what is marked

Vague words to replace

These everyday words are the commonest reason a correct idea scores nothing in Chemistry. Swap each one for the scientific wording.

Vague word or phrase	Write this instead
"stronger" or "weaker" with no noun	Name what is stronger: "stronger London forces", "stronger electrostatic attraction", "stronger metallic bonding"
"bonds" for everything	Say "covalent bonds" (within molecules), "ionic bonding", "hydrogen bonds" or "London forces" (between molecules)
"molecules break up" when boiling a simple molecular substance	"Intermolecular forces are overcome; the covalent bonds are not broken"
"it" or "the chemical"	Name the species: "the hydroxide ion", "the carbocation", "the Mg^{2+} ion"
"attracted to the nucleus more"	"Greater nuclear charge, same shielding, so the outer electrons are held more strongly"
"gets used up" for catalysts	"Is not used up; it provides an alternative route with a lower activation energy"
"the reaction goes faster"	"More frequent collisions with energy greater than or equal to the activation energy, so more successful collisions per unit time"
"the equilibrium goes to the right"	"The position of equilibrium shifts to the right to oppose the change"
"heat" and "temperature" used as the same thing	"Heat" is energy transferred; "temperature" is a measure of mean kinetic energy
"energy is lost"	"Energy is transferred to the surroundings" (conservation of energy)
"atoms" for ions or molecules	Use "ions" for charged particles, "molecules" for covalent substances, "atoms" only for uncombined atoms
"the electrons move" in a mechanism	"A lone pair on the nucleophile attacks the delta-plus carbon"
"a bit", "around", "about the same"	Give a value with units, or say "greater than", "less than", "equal to"
"pure" or "clean"	"Free from impurities"; or give the measured evidence such as a sharp melting point

Vague word or phrase	Write this instead
"colour goes away"	"The solution is decolourised" or state the colour change from X to Y
"acid reacts with base"	"The H+ ions react with OH- ions to form water"

Sentence frames to reuse

Frame	Pattern	Example
Trend explanation	Going [down or across], [size or charge] [increases], so [attraction] [weakens or strengthens], so [property] [changes].	Going down Group 2, the atomic radius and shielding increase, so the outer electrons are held less strongly, so the first ionisation energy decreases.
Property from structure	[Substance] has [structure], with [bond or force] between [particles]; a lot of energy is needed to overcome these, so [property].	Magnesium oxide has a giant ionic lattice with strong electrostatic attraction between ions; a lot of energy is needed to overcome this, so it has a high melting point.
Mechanism	A lone pair on [nucleophile] attacks the delta-plus [atom]; the [bond] breaks and the electrons go to [atom].	A lone pair on the hydroxide ion attacks the delta-plus carbon; the C-Br bond breaks and both electrons go to bromine.
Equilibrium	If [change], the system opposes it, so the position of equilibrium shifts [direction], which [effect on yield].	If the pressure increases, the position shifts to the side with fewer moles of gas, which increases the yield.
Collision theory	[Change] gives [more frequent collisions or a greater proportion above Ea], so more successful collisions per unit time, so the rate increases.	A higher temperature gives a greater proportion of molecules with energy at or above the activation energy.
Calculation	Formula, substitution with units, answer to the correct significant figures, with units.	n = c x V = 0.100 x 0.02245 = 2.245 x 10-3 mol

Topic 1: Atomic Structure and the Periodic Table

Concept	Own words (loses marks)	Exam language (scores)
Isotopes	Same element, different weight.	Atoms of the same element with the same number of protons but different numbers of neutrons.
Relative atomic mass	The average mass of an atom.	The weighted mean mass of an atom of an element compared with one twelfth of the mass of an atom of carbon-12.
Calculating Ar from mass spectrum data	Add the masses and divide.	Ar = the sum of (isotope mass multiplied by its abundance) divided by the total abundance.
Electron configuration	Electrons go in shells.	Electrons fill the lowest energy subshells first. Each orbital holds a maximum of two electrons with opposite spins, and orbitals of the same subshell are filled singly before they are paired (Hund's rule).
Chromium and copper	Chromium is an exception.	Chromium is [Ar] 3d5 4s1 and copper is [Ar] 3d10 4s1, because a half-filled or completely filled 3d subshell is more stable.
First ionisation energy	The energy to remove an electron.	The energy required to remove one electron from each atom in one mole of gaseous atoms to form one mole of gaseous 1+ ions.
Trend down a group	It gets easier to remove an electron.	Down a group the outer electron is further from the nucleus and there is more shielding, so the attraction is weaker and the first ionisation energy decreases.

Concept	Own words (loses marks)	Exam language (scores)
Trend across a period	It gets harder to remove an electron.	Across a period the nuclear charge increases with similar shielding, so the atomic radius decreases and the outer electron is held more strongly, so the first ionisation energy generally increases.
Dip from Mg to Al	Aluminium is different.	The outer electron of aluminium is in a 3p subshell, which is higher in energy than the 3s subshell and shielded by the 3s electrons, so less energy is needed to remove it.
Dip from P to S	Sulfur has paired electrons.	In sulfur the outer 3p electron is paired with another in the same orbital, so electron-electron repulsion makes it easier to remove than the unpaired electron in phosphorus.
Large jump in successive ionisation energies	There is a big jump.	A large increase shows the next electron is being removed from an inner shell, which is closer to the nucleus and less shielded. The number of electrons removed before the jump equals the number of outer electrons, which gives the group.
Melting points across Period 3	Metals melt high and gases melt low.	Sodium to aluminium have metallic bonding that strengthens as ion charge increases and ion size decreases. Silicon is giant covalent, so many strong covalent bonds must be broken. P4, S8 and Cl2 are simple molecules with weak London forces between them, so they have low melting points.

Key terms to include: isotope, relative atomic mass, weighted mean, subshell, orbital, Hund's rule, shielding, nuclear charge, successive ionisation energy, gaseous, 1+ ion.

Topic 2: Bonding and Structure

Concept	Own words (loses marks)	Exam language (scores)
Ionic bonding	A metal gives electrons to a non-metal.	The electrostatic attraction between oppositely charged ions in a giant ionic lattice.
Properties of ionic compounds	They conduct when dissolved.	They have high melting points because a lot of energy is needed to overcome the strong electrostatic attraction. They do not conduct when solid because the ions are fixed in position, but conduct when molten or aqueous because the ions are free to move.
Covalent bond	Atoms share electrons.	The electrostatic attraction between a shared pair of electrons and the nuclei of the two bonded atoms.
Dative covalent bond	Both electrons come from one atom.	A shared pair of electrons in which both electrons are supplied by one atom, which donates a lone pair to an electron-deficient atom or ion, for example in NH4+.
Metallic bonding	A sea of electrons.	The electrostatic attraction between positive metal ions and delocalised electrons. Metals conduct because the delocalised electrons carry charge, and are malleable because the layers of ions slide while the bonding is maintained.

Concept	Own words (loses marks)	Exam language (scores)
VSEPR and lone pairs	The pairs push each other away.	Electron pairs repel to be as far apart as possible. A lone pair repels more than a bonding pair, so bond angles are reduced: CH4 109.5 degrees, NH3 107 degrees, H2O 104.5 degrees.
Molecular shapes	It's a pyramid.	Four bonding pairs: tetrahedral, 109.5 degrees. Three bonding pairs and one lone pair: trigonal pyramidal, 107 degrees. Two bonding pairs and two lone pairs: bent, 104.5 degrees. Three bonding pairs: trigonal planar, 120 degrees. Five: trigonal bipyramidal, 90 and 120 degrees. Six: octahedral, 90 degrees.
Polar bonds and polar molecules	Unequal sharing makes it polar.	A difference in electronegativity gives a polar bond with delta-plus and delta-minus ends. A molecule is polar only if the dipoles do not cancel by symmetry, so CO2 is non-polar although its C=O bonds are polar.
London forces	Weak forces between molecules.	Instantaneous dipole-induced dipole forces. Larger molecules with more electrons have stronger London forces, so more energy is needed to overcome them and the boiling point is higher.
Hydrogen bonding	A strong special bond.	An intermolecular force between a lone pair on N, O or F and a delta-plus hydrogen atom bonded to N, O or F. It explains the high boiling point of water, and ice is less dense because hydrogen bonds hold the molecules in an open lattice.

Concept	Own words (loses marks)	Exam language (scores)
Giant covalent structures	Diamond is hard and graphite conducts.	In diamond each carbon is bonded to four others by strong covalent bonds, so it is hard and has no free electrons. In graphite each carbon is bonded to three others in layers with delocalised electrons that carry charge, and weak London forces between the layers let them slide.
Simple molecular substances	They have low melting points.	Weak intermolecular forces are overcome, which needs little energy; the covalent bonds within the molecules are not broken. They do not conduct because there are no charged particles that are free to move.

Key terms to include: electrostatic attraction, giant ionic lattice, delocalised electrons, lone pair, electronegativity, dipole, London forces, hydrogen bond, intermolecular, shielding, giant covalent.

Topic 3: Redox I

Concept	Own words (loses marks)	Exam language (scores)
Oxidation and reduction	Oxidation is adding oxygen.	Oxidation is the loss of electrons, or an increase in oxidation number. Reduction is the gain of electrons, or a decrease in oxidation number.
Oxidation number rules	Work out the numbers.	An uncombined element is 0. A simple ion has its charge. Oxygen is -2 (except in peroxides, -1) and hydrogen is +1 (except in metal hydrides, -1). The sum equals the overall charge.

Concept	Own words (loses marks)	Exam language (scores)
Oxidising and reducing agents	The one that is oxidised.	An oxidising agent accepts electrons and is itself reduced. A reducing agent donates electrons and is itself oxidised.
Disproportionation	A substance reacts with itself.	The same element is both oxidised and reduced in one reaction. In Cl2 + 2NaOH giving NaCl + NaClO + H2O, chlorine goes from 0 to -1 (reduced) and to +1 (oxidised).
Balancing a half-equation	Balance the equation.	Balance the atoms, then balance oxygen with H2O, hydrogen with H+, and charge with electrons. Combine two half-equations so the electrons cancel.
Dichromate and permanganate	It changes colour.	Cr2O7 2- + 14H+ + 6e- giving 2Cr3+ + 7H2O: orange to green. MnO4- + 8H+ + 5e- giving Mn2+ + 4H2O: purple to colourless (or very pale pink).

Topic 4: Inorganic Chemistry and the Periodic Table

Concept	Own words (loses marks)	Exam language (scores)
Reactivity down Group 2	They get more reactive.	Down the group the atomic radius and shielding increase, so the outer two electrons are lost more easily, the ionisation energies are lower, and reactivity increases.

Concept	Own words (loses marks)	Exam language (scores)
Solubility of Group 2 hydroxides and sulfates	Some dissolve and some don't.	Hydroxide solubility increases down the group, so Mg(OH)2 is the least soluble. Sulfate solubility decreases down the group, so BaSO4 is the least soluble; a white precipitate with barium chloride in acid shows sulfate ions are present.
Thermal stability of carbonates	They break down more easily at the top.	Down the group the cations get larger, so the charge density decreases. The carbonate ion is polarised less, so more energy is needed to decompose it and thermal stability increases.
Trends in Group 7	They get less reactive.	Down the group the boiling point increases because of stronger London forces. The oxidising power decreases because the atoms are larger with more shielding, so an incoming electron is attracted less strongly.
Halogen displacement	Chlorine pushes out bromine.	Chlorine is a stronger oxidising agent than bromine, so it oxidises bromide ions: Cl2 + 2Br- giving 2Cl- + Br2. The solution turns orange.
Halide ion test	Add silver nitrate.	Add dilute nitric acid, then silver nitrate solution. Chloride gives a white precipitate, bromide a cream precipitate and iodide a yellow precipitate. AgCl dissolves in dilute ammonia, AgBr in concentrated ammonia, and AgI does not dissolve.

Concept	Own words (loses marks)	Exam language (scores)
Halides with concentrated sulfuric acid	Fumes come off.	Chloride gives steamy white fumes of HCl only. Bromide gives HBr, then some is oxidised to orange-brown Br2 and SO2. Iodide is a stronger reducing agent, so it reduces the acid to SO2, sulfur and hydrogen sulfide, which smells of bad eggs.
Chlorine in water and with alkali	Chlorine kills bacteria.	Chlorine disproportionates in water to give HCl and chloric(I) acid, HClO, which kills bacteria. With cold dilute sodium hydroxide it forms NaCl and NaClO (bleach).
Other anion and cation tests	Test for a gas.	Carbonate: add acid, CO2 turns limewater milky. Sulfate: barium chloride and HCl, white precipitate. Ammonium: warm with NaOH, ammonia turns damp red litmus blue.

Topic 5: Formulae, Equations and Amounts of Substance

Concept	Own words (loses marks)	Exam language (scores)
The mole	An amount of stuff.	The amount of substance that contains 6.02 x 10^23 particles of the stated type (the Avogadro constant).
Mole formulae	Use the right equation.	n = mass divided by Mr; n = concentration (mol dm-3) x volume (dm3); n = gas volume divided by 24 dm3 mol-1 at room temperature and pressure. Convert cm3 to dm3 by dividing by 1000.

Concept	Own words (loses marks)	Exam language (scores)
Ideal gas equation	pV = nRT.	pV = nRT with p in Pa, V in m3, T in kelvin and R = 8.31 J K-1 mol-1. Convert cm3 to m3 by dividing by 10^6 and kPa to Pa by multiplying by 1000.
Empirical and molecular formula	The simplest formula.	Divide each mass by its Ar to get moles, divide by the smallest to get the ratio, and scale to whole numbers. The molecular formula is the empirical formula multiplied by Mr divided by the empirical formula mass.
Percentage yield	How much I made.	Percentage yield = actual yield divided by theoretical yield, multiplied by 100. The theoretical yield is from the limiting reagent.
Atom economy	The waste.	Atom economy = Mr of the desired product divided by the sum of the Mr of all products (or reactants), multiplied by 100.
Limiting reagent	The one that runs out.	Calculate the moles of each reactant, divide each by its coefficient in the equation, and the smallest value is the limiting reagent.
Titration calculation	Use the titre to work it out.	Use the mean of concordant titres (within 0.10 cm3). Calculate the moles of the known solution, use the mole ratio from the equation to get the moles of the unknown, then divide by its volume (and scale up for any dilution).
Significant figures	Round the answer.	Give the final answer to the same number of significant figures as the least precise data, and do not round intermediate values.

Key terms to include: oxidation number, electrons, disproportionation, charge density, polarised, oxidising agent, limiting reagent, mole ratio, concordant, theoretical yield, atom economy.

Topic 6: Organic Chemistry I

Concept	Own words (loses marks)	Exam language (scores)
Homologous series	A family of compounds.	A series of compounds with the same functional group and the same general formula, each member differing from the next by CH2, with similar chemical properties and gradually changing physical properties.
Structural isomers and stereoisomers	Same atoms in a different order.	Structural isomers have the same molecular formula but different structural formulae. Stereoisomers have the same structural formula but a different arrangement of atoms in space.
E/Z isomerism	The groups are on the same or different sides.	It occurs because rotation about the C=C double bond is restricted, and each carbon of the double bond is bonded to two different groups. Z has the higher-priority groups on the same side; E has them on opposite sides.
Cracking	Big molecules are broken up.	Cracking breaks long-chain alkanes into shorter alkanes and alkenes, either thermally (high temperature and pressure) or catalytically (a zeolite catalyst at about 450 degrees C).

Concept	Own words (loses marks)	Exam language (scores)
Free radical substitution	Chlorine reacts with methane.	Initiation: UV light breaks Cl2 into two chlorine radicals. Propagation: Cl radical + CH4 gives HCl + methyl radical, then methyl radical + Cl2 gives CH3Cl + Cl radical. Termination: two radicals combine. Further substitution gives a mixture of products.
Electrophilic addition	HBr adds across the double bond.	The pi electrons of the C=C attack the delta-plus hydrogen of HBr (arrow from the double bond to H, and from the H-Br bond to Br), forming a carbocation. A lone pair on Br- then attacks the C+ to form the product.
Major product of addition	The right-hand side gets the bromine.	The more substituted carbocation (secondary rather than primary) is more stable, because alkyl groups push electron density towards the positive carbon, so it forms preferentially and gives the major product.
Test for an alkene	The colour goes away.	Shake with orange bromine water; an alkene decolourises it (orange to colourless).
Nucleophilic substitution	Hydroxide replaces bromine.	A lone pair on the hydroxide ion (a nucleophile) attacks the delta-plus carbon; the C-Br bond breaks and both electrons go to bromine. Use aqueous NaOH and warm under reflux.

Concept	Own words (loses marks)	Exam language (scores)
Elimination	Hydrogen bromide is taken out.	Heat with ethanolic potassium hydroxide; the hydroxide ion acts as a base, removing a hydrogen from the carbon next to the C-Br, and the alkene is formed.
Rate of hydrolysis of halogenoalkanes	Iodine reacts fastest.	The C-I bond is the weakest, so it is broken most easily and the iodoalkane reacts fastest. Warm with aqueous silver nitrate in ethanol and time the precipitate: AgI is yellow, AgBr cream, AgCl white.
Oxidation of alcohols	Alcohols turn into other things.	Primary alcohols are oxidised to aldehydes (distil off as they form) then to carboxylic acids (reflux with excess oxidising agent). Secondary alcohols give ketones. Tertiary alcohols are not oxidised. Acidified potassium dichromate changes from orange to green.
Reflux and distillation	Heat the mixture.	Reflux: the vapour condenses and returns to the flask, so volatile reactants and products are not lost and heating can continue. Distillation: the product is removed as it forms, to prevent further oxidation.
Dehydration of an alcohol	Water comes off the alcohol.	Heat the alcohol with concentrated sulfuric acid or phosphoric acid (or pass the vapour over hot aluminium oxide) to form an alkene and water.

Topic 7: Modern Analytical Techniques I

Concept	Own words (loses marks)	Exam language (scores)
Molecular ion peak	The tallest peak.	The peak with the highest m/z is the molecular ion, M+, and its m/z equals the relative molecular mass. The tallest peak is the base peak, which is the most abundant ion.
Fragmentation	The molecule breaks up.	The molecular ion fragments into a positive ion, which is detected, and a radical, which is not. Common fragments are 15 (CH_3^+), 29 ($C_2H_5^+$), 43 ($C_3H_7^+$ or CH_3CO^+) and 57 ($C_4H_9^+$).
Infrared spectroscopy	Different bonds show different peaks.	Bonds absorb infrared radiation at characteristic wavenumbers, which makes the bonds vibrate. O-H in an alcohol is broad at 3200 to 3550 cm^{-1}; O-H in a carboxylic acid is very broad at 2500 to 3300; C=O is a strong sharp peak at 1680 to 1750; C-H is at about 2850 to 3100.
Fingerprint region	The bottom part.	The region below about 1500 cm^{-1}, unique to each compound, so it can be matched to a database to identify a compound.
Combining techniques	Use them all.	Use the mass spectrum for Mr and fragments, the infrared spectrum for functional groups, and the molecular formula and fragments to propose a structure; check that every piece of data fits it.

Key terms to include: functional group, general formula, structural isomer, stereoisomer, restricted rotation, radical, initiation, propagation, termination, carbocation, nucleophile, electrophile, reflux, distillation, molecular ion, fragment, wavenumber, fingerprint region.

Topic 8: Energetics I

Concept	Own words (loses marks)	Exam language (scores)
Exothermic and endothermic	One gives out heat and one takes it in.	In an exothermic reaction energy is transferred to the surroundings, the enthalpy change is negative and the temperature of the surroundings rises. In an endothermic reaction energy is taken in from the surroundings, the enthalpy change is positive and the temperature falls.
Standard conditions	Normal conditions.	A pressure of 100 kPa, a stated temperature (usually 298 K), and every substance in its standard state.
Enthalpy of formation	Making a compound.	The enthalpy change when one mole of a compound is formed from its elements in their standard states under standard conditions.
Enthalpy of combustion	Burning something.	The enthalpy change when one mole of a substance is completely burned in oxygen under standard conditions.
Hess's law	The route doesn't matter.	The enthalpy change of a reaction is independent of the route taken, provided the initial and final conditions are the same.
Calorimetry calculation	Use the temperature rise.	q = m x c x change in temperature, where m is the mass of water in grams and c = 4.18 J g-1 K-1. Convert q to kJ, divide by the moles reacted, and add the sign: negative if the temperature rose.

Concept	Own words (loses marks)	Exam language (scores)
Reasons for a calorimetry error	Heat gets lost.	Energy is transferred to the surroundings, combustion is incomplete, the specific heat capacity of the solution is assumed to equal that of water, and the standard conditions are not met. Use a lid and insulation to reduce heat loss.
Mean bond enthalpy	The energy in a bond.	The average energy needed to break one mole of a given bond in the gaseous state, averaged over a range of compounds. Enthalpy change = bonds broken minus bonds formed. The values are means, so the answer is only approximate.

Topic 9: Kinetics I

Concept	Own words (loses marks)	Exam language (scores)
Collision theory	They have to collide.	A reaction occurs only when particles collide with energy greater than or equal to the activation energy and with the correct orientation.
Activation energy	The energy to start the reaction.	The minimum energy that colliding particles must have in order to react.
Concentration and pressure	More particles means more reactions.	More particles in the same volume give more frequent collisions, so more successful collisions per unit time and a higher rate.
Surface area	Powder reacts faster.	A larger surface area exposes more particles to collision, so there are more frequent collisions and the rate increases.

Concept	Own words (loses marks)	Exam language (scores)
Temperature	Heat makes the particles faster.	A higher temperature increases the mean kinetic energy, so a greater proportion of molecules have energy greater than or equal to the activation energy, and collisions are more frequent. The first effect has the greater influence.
Maxwell-Boltzmann distribution	The curve shifts.	The area under the curve is the total number of molecules and stays the same. At a higher temperature the peak is lower and moves to the right, so a greater proportion of molecules are to the right of the activation energy.
Catalyst	It speeds it up and isn't used up.	A catalyst provides an alternative reaction route with a lower activation energy, so at the same temperature a greater proportion of collisions are successful. It is not used up, and it does not change the enthalpy change or the position of equilibrium.

Topic 10: Equilibrium I

Concept	Own words (loses marks)	Exam language (scores)
Dynamic equilibrium	Both reactions happen equally.	In a closed system the rate of the forward reaction equals the rate of the reverse reaction, so the concentrations of reactants and products stay constant.
Le Chatelier's principle	The equilibrium fights back.	If a change is made to a system at equilibrium, the position of equilibrium shifts in the direction that opposes the change.

Concept	Own words (loses marks)	Exam language (scores)
Effect of concentration	It moves to the other side.	Adding a reactant shifts the position to the right, using up some of the added reactant; removing a product also shifts it to the right.
Effect of pressure	High pressure favours the product.	Increasing the pressure shifts the position towards the side with fewer moles of gas, which reduces the pressure. If the moles of gas are equal on both sides there is no shift.
Effect of temperature	Heat favours one side.	Increasing the temperature shifts the position in the endothermic direction, which absorbs heat. For an exothermic forward reaction the yield falls.
Effect of a catalyst	It improves the yield.	A catalyst increases the rate of both the forward and reverse reactions equally, so equilibrium is reached faster but the position and the value of Kc do not change.
Kc expression and units	Products over reactants.	Kc is the equilibrium concentration of the products, each raised to the power of its coefficient, divided by that of the reactants. Work out the units by cancelling the concentration units in the expression.
Kc and temperature	Kc changes with conditions.	Kc changes only when the temperature changes. Concentration, pressure and a catalyst do not change Kc.
Industrial compromise	They use a medium temperature.	A lower temperature increases the equilibrium yield of an exothermic reaction, but the rate is too slow. A compromise temperature gives an acceptable rate and yield, for example 450 degrees C for the Haber process with an iron catalyst and 200 atm.

Key terms to include: exothermic, standard state, Hess's law, mean bond enthalpy, activation energy, orientation, proportion of molecules, Maxwell-Boltzmann, alternative route, dynamic equilibrium, position of equilibrium, Le Chatelier, closed system.

Topic 13: Energetics II

This is the newest and highest-risk content. Learn these definitions word for word.

Concept	Own words (loses marks)	Exam language (scores)
Lattice enthalpy of formation	The energy released when a lattice forms.	The enthalpy change when one mole of an ionic solid is formed from its gaseous ions under standard conditions. It is always exothermic.
Lattice enthalpy of dissociation	The energy to break up a lattice.	The enthalpy change when one mole of an ionic solid is broken up into its gaseous ions. It is endothermic and has the same magnitude as the lattice enthalpy of formation.
Born-Haber cycle	Add up all the steps.	Hess's law applied to ionic compounds: enthalpy of formation = atomisation of the metal + ionisation energies + atomisation of the non-metal + electron affinities + lattice enthalpy of formation. Multiply the non-metal steps by the number of atoms in the formula.
Electron affinity	The energy when an electron is added.	The enthalpy change when one mole of gaseous atoms gains one mole of electrons to form one mole of gaseous 1- ions. The first is exothermic; the second, for example O- to O2-, is endothermic because the electron is added to a negative ion.

Concept	Own words (loses marks)	Exam language (scores)
Factors affecting lattice enthalpy	Bigger ions are weaker.	The lattice enthalpy of formation is more exothermic when the ionic charge is greater and the ionic radius is smaller, because the electrostatic attraction between the ions is stronger. MgO is more exothermic than NaCl.
Theoretical and experimental values	The values are different.	The theoretical value assumes a purely ionic model with spherical ions. The experimental value is more exothermic because of covalent character, where a small, highly charged cation polarises the anion and adds extra bonding.
Enthalpy of hydration and solution	Dissolving a salt.	Enthalpy of hydration: the enthalpy change when one mole of gaseous ions forms aqueous ions. Enthalpy of solution = lattice enthalpy of dissociation + the enthalpies of hydration of the ions.
Entropy	Disorder.	A measure of the number of ways that particles and energy can be arranged. Gases have a higher entropy than liquids, and liquids higher than solids. Entropy increases when a gas is produced or a solid dissolves.
Entropy change of a reaction	Products minus reactants.	The entropy change of the system = the sum of the entropies of the products minus the sum of the entropies of the reactants. The entropy change of the surroundings = minus the enthalpy change divided by the temperature (in kelvin).

Concept	Own words (loses marks)	Exam language (scores)
Free energy and feasibility	It will happen.	The Gibbs free energy change = enthalpy change minus (temperature x entropy change). A reaction is feasible when it is zero or negative. The minimum temperature is found by setting it equal to zero: T = enthalpy change divided by entropy change. Convert kJ to J first.
Feasible but not happening	The reaction is slow.	A reaction can be feasible but not occur because the activation energy is too high, so the rate is too slow. Feasibility says nothing about the rate.

Key terms to include: gaseous ions, one mole, standard conditions, exothermic, endothermic, electron affinity, charge, ionic radius, polarise, covalent character, hydration, entropy, feasible, Gibbs free energy.

Topic 17 (sections A and B): Chirality and Carbonyl Compounds

Concept	Own words (loses marks)	Exam language (scores)
Chiral centre	A carbon with four different things.	A carbon atom bonded to four different atoms or groups. The molecule has a non-superimposable mirror image, which gives a pair of enantiomers.
Optical activity	It rotates the light.	Enantiomers rotate the plane of plane-polarised light in opposite directions by equal amounts. One is dextrorotatory (+) and the other is laevorotatory (-).
Racemic mixture	A mixture of both.	An equimolar mixture of two enantiomers. It is optically inactive because the rotations of the two enantiomers cancel out.

Concept	Own words (loses marks)	Exam language (scores)
Polarity of the carbonyl group	The oxygen is greedy.	Oxygen is more electronegative than carbon, so the C=O bond is polar with a delta-plus carbon and a delta-minus oxygen. The delta-plus carbon is attacked by nucleophiles.
Nucleophilic addition with HCN	Cyanide adds to the carbonyl.	A lone pair on the carbon of CN- attacks the delta-plus carbon (arrow from the lone pair to C). The C=O bond breaks, with the electrons going to oxygen (arrow from the bond to O). A lone pair on O- then takes H+ from HCN (or water) to give the hydroxynitrile.
Why the product is racemic	It forms a mixture.	The carbonyl group is planar, so the nucleophile can attack from either side with equal probability. This forms equal amounts of both enantiomers, and the mixture does not rotate polarised light.
Reduction of carbonyls	Add hydrogen.	NaBH4 in aqueous solution (or LiAlH4 in dry ether) acts as a source of hydride ions, H-, which are nucleophiles. Aldehydes are reduced to primary alcohols and ketones to secondary alcohols.
Distinguishing an aldehyde from a ketone	Do a test.	Warm with Tollens' reagent: an aldehyde forms a silver mirror and a ketone does not. Alternatively, warm with acidified potassium dichromate: an aldehyde turns it from orange to green and a ketone does not change.

Concept	Own words (loses marks)	Exam language (scores)
Test for a carbonyl group	The solution changes colour.	Add 2,4-dinitrophenylhydrazine (2,4-DNPH): an orange or yellow precipitate shows an aldehyde or ketone. Filter, recrystallise and measure the melting point, then compare it with a database to identify the compound.
Why chirality matters in medicines	It's important in drugs.	Enantiomers can have different biological effects, because enzymes and receptors are chiral and one enantiomer fits better. A single enantiomer is often needed, which makes it important to make or separate pure enantiomers.

Key terms to include: chiral centre, enantiomer, non-superimposable, plane-polarised light, racemic mixture, equimolar, optically inactive, delta-plus, nucleophilic addition, hydroxynitrile, hydride ion, Tollens' reagent, 2,4-DNPH.

Topic 19: Modern Analytical Techniques II

Concept	Own words (loses marks)	Exam language (scores)
What NMR shows	It tells you about the hydrogens.	NMR uses radio waves to detect nuclei in a magnetic field. Each peak is a different environment of hydrogen (or carbon) atoms in the molecule.
Chemical shift	Where the peak is.	The position of a peak in ppm relative to TMS, which is at 0 ppm. It depends on the chemical environment: nearby electronegative atoms or groups move the peak to a higher shift.

Concept	Own words (loses marks)	Exam language (scores)
Number of peaks	How many hydrogens.	The number of peaks equals the number of different environments, not the number of hydrogens. In 13C NMR it equals the number of different carbon environments.
Integration	The size of the peak.	The area under each peak is proportional to the number of hydrogen atoms in that environment, which gives the ratio of hydrogens.
Splitting pattern	The peak splits.	The n + 1 rule: a signal is split into n + 1 peaks by n hydrogens on the adjacent carbon. A quartet means three hydrogens on the adjacent carbon, a triplet two, a doublet one, and a singlet none.
OH and exchange with D2O	The OH peak goes away.	Shaking with D2O replaces the OH hydrogen with deuterium, which does not appear in the spectrum, so the OH peak disappears. This identifies the OH peak.
Why TMS is used	It is the standard.	TMS gives a single, sharp peak because its 12 hydrogens are all equivalent. It is inert, volatile (easily removed), and its peak is at a shift away from most others.
Solvents in proton NMR	The solvent is a liquid.	A deuterated solvent such as CDCl3 or D2O is used, or a solvent with no hydrogen such as CCl4, so the solvent gives no proton signals that would mask the sample's peaks.
Thin-layer chromatography	The spots separate.	Rf = distance moved by the spot divided by distance moved by the solvent front. Components that are more soluble in the mobile phase, or adsorbed less strongly on the stationary phase, move further.

Concept	Own words (loses marks)	Exam language (scores)
Gas chromatography and GC-MS	It separates a mixture.	Components are separated by their different retention times, depending on how strongly each interacts with the stationary phase. In GC-MS, each separated component is then passed into a mass spectrometer, which gives its molecular ion and fragments for identification.
Deducing a structure	Look at everything.	Use the mass spectrum for Mr, the infrared spectrum for functional groups, the carbon-13 NMR for the number of carbon environments, and the proton NMR for the environments, ratio and neighbours. Check that the proposed structure matches every piece of data.

Key terms to include: chemical shift, TMS, environment, integration, splitting, n + 1 rule, deuterated, retention time, mobile phase, stationary phase, Rf.

Skills: calculations, practicals, errors, mechanisms and long answers

Calculations

Situation	Own words (loses marks)	Exam language (scores)
Showing working	Write the answer.	Write the formula, substitute the values with units, then give the answer with units and to the correct significant figures. A wrong answer with correct working still earns the method marks.

Situation	Own words (loses marks)	Exam language (scores)
Units	Forget the unit or use the wrong one.	Give units on every final answer: mol, g, mol dm-3, kJ mol-1, J K-1 mol-1. Convert J and kJ before combining a free energy calculation.
Mole ratio	Use the numbers from the equation.	Use the coefficients in the balanced equation as the mole ratio, not the masses.
Percentage uncertainty	The error as a percentage.	Percentage uncertainty = (uncertainty divided by the measured value) x 100. A burette reading has an uncertainty of +/-0.05 cm3, and a titre uses two readings, so the total is +/-0.10 cm3.
Combining uncertainties	Add them up.	When quantities are added or subtracted, add the absolute uncertainties. When they are multiplied or divided, add the percentage uncertainties.
Percentage difference from a data book value	How far off I was.	Percentage difference = (measured value minus accepted value) divided by the accepted value, x 100.

Practical techniques

Technique	Own words (loses marks)	Exam language (scores)
Titration technique	Add the acid slowly and see when it changes.	Rinse the burette with the solution it will contain, remove the funnel, read the bottom of the meniscus at eye level, swirl, add dropwise near the end point, and use a white tile. Repeat until two titres are concordant (within 0.10 cm3).

Technique	Own words (loses marks)	Exam language (scores)
Why do a rough titration	To practise.	A rough titre gives the approximate volume of the end point, so the accurate titrations can add the solution dropwise as the end point is approached.
Making a standard solution	Mix the solid with water.	Weigh by difference, dissolve the solid in a beaker with distilled water, transfer it to a volumetric flask rinsing the beaker and funnel into the flask, make up to the mark with the bottom of the meniscus on the line, and invert to mix.
Reflux apparatus	Heat the flask.	The mixture is heated in a flask fitted with a vertical condenser, so vapours condense and return to the flask. Add anti-bumping granules, and do not seal the top.
Distillation apparatus	Collect the liquid.	Heat the mixture and collect the fraction that boils at the boiling point of the product, with the thermometer bulb level with the side arm of the condenser and the water flowing in at the bottom of the condenser.
Measuring a rate	Time how long it takes.	Measure the volume of gas collected at set times, or the loss in mass. The initial rate is found from the gradient of the tangent at time zero.
Choosing an indicator	Any indicator works.	For a weak acid and a strong base use phenolphthalein (colourless to pink); for a strong acid and a weak base use methyl orange (red to yellow).

Errors and improving a method

Idea	Exam language (scores)
Random and systematic errors	Random errors vary from reading to reading and are reduced by repeating and averaging. A systematic error shifts all the results in the same direction and is reduced by improving the method or the instrument.
Accuracy and precision	Accuracy is how close a result is to the true value; precision is how close repeated results are to each other.
Improving calorimetry	Use a lid and insulation to reduce heat loss, stir to give an even temperature, and use a more precise thermometer.
Improving a titration	Use a smaller volume of the more concentrated solution, or a more precise burette, so the percentage uncertainty is smaller.
Practical yield	A yield below 100 per cent is due to incomplete reaction, transfer losses, side reactions, and loss during purification or recrystallisation.

Mechanism rules (curly arrows)

Rule	What to do
Arrow start	A curly arrow starts from a lone pair or a bond (or a negative charge), never from an atom.
Arrow end	The head points to the atom or bond that receives the electron pair.
Show the dipoles	Mark delta-plus and delta-minus on polar bonds, and show lone pairs on nucleophiles.
Intermediates	Draw the carbocation or the alkoxide intermediate in brackets with its charge.
Radicals	A single-barbed (fishhook) arrow for each electron, and a dot for the unpaired electron.

Rule	What to do
Name the type	State "nucleophilic substitution", "electrophilic addition", "nucleophilic addition" or "free radical substitution".

Six-mark and extended answers

1. Spend one minute listing six or more separate points before writing. Each is a possible mark.
2. Open with a sentence that answers the question directly.
3. Write one point per sentence using key terms and a link such as "so" or "which means".
4. For a structure question, state what each piece of data shows, then the structure, then check it against all the data.
5. For an evaluation, give an advantage and a disadvantage of each option, then a justified judgement.
6. Do not repeat a point in new words; the mark scheme gives credit once.

Practice routine and checklist

The three-pass routine (15 minutes a day, one topic block at a time)

Pass	What to do	Time
1. Recall	Read the concept, cover the answer column and write the answer from memory in full sentences	5 minutes
2. Compare	Uncover the exam-language answer, then underline every key term that was missing or vague in the written answer	4 minutes
3. Rewrite	Cover the page again and rewrite the answer using the missing terms; speak it aloud once	6 minutes

Work through the topics in this order: Topic 13 first (newest and highest risk), then Topics 17 and 19, then Topics 2, 4 and 10, then the rest. Use the flashcard deck on a phone for short gaps in the day.

Self-check before handing in any answer

- [ ] Does every definition contain all its key phrases (one mole, gaseous, standard conditions)?

- [ ] Have I named the force or bond, and said whether it is within or between molecules?
- [ ] Does each "explain" answer give a chain: size or charge, attraction, energy needed, property?
- [ ] Do all mechanism arrows start on a lone pair or a bond?
- [ ] For calculations, have I shown the formula, the substitution and the units?
- [ ] For structure questions, does the structure fit every piece of data?

Progress tracker

After each practice paper, count the marks lost to "imprecise wording" using the diagnostic grid in the mock paper's mark scheme. The aim is to see that number fall paper by paper.